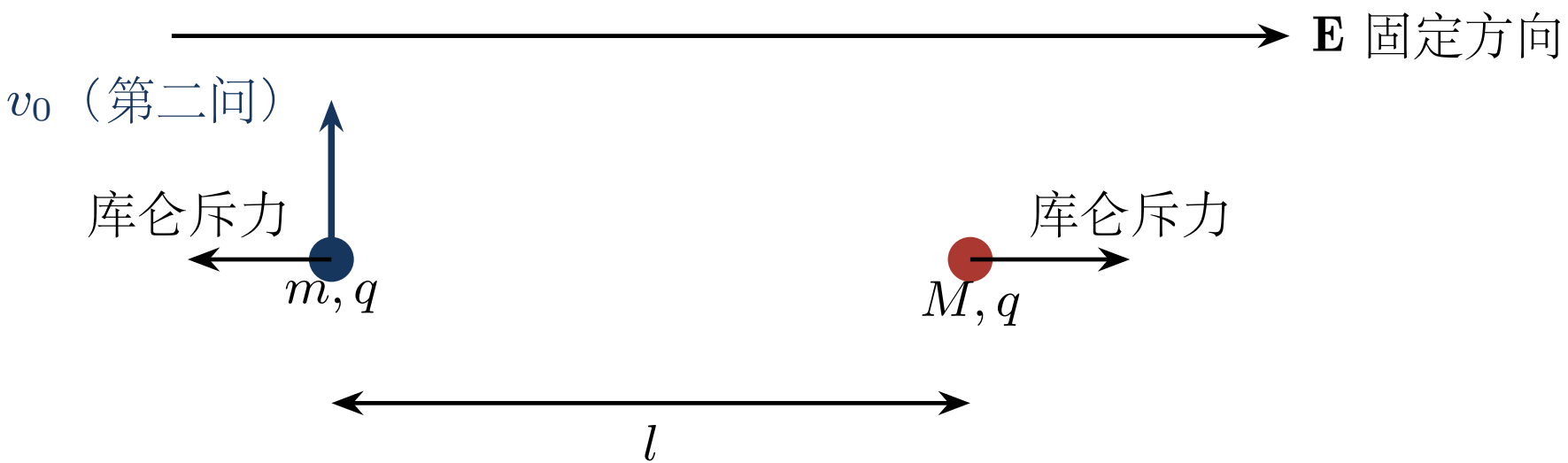
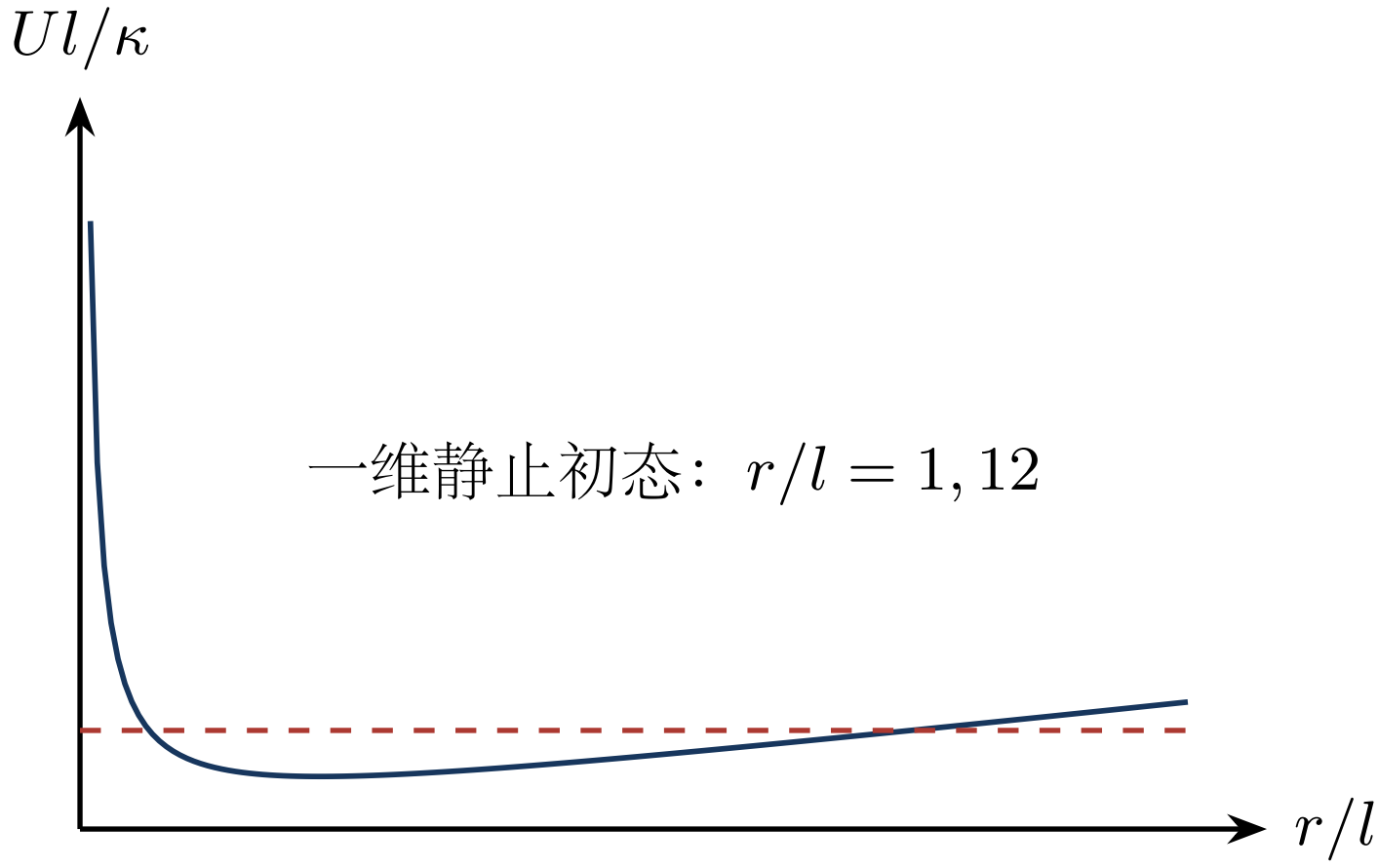


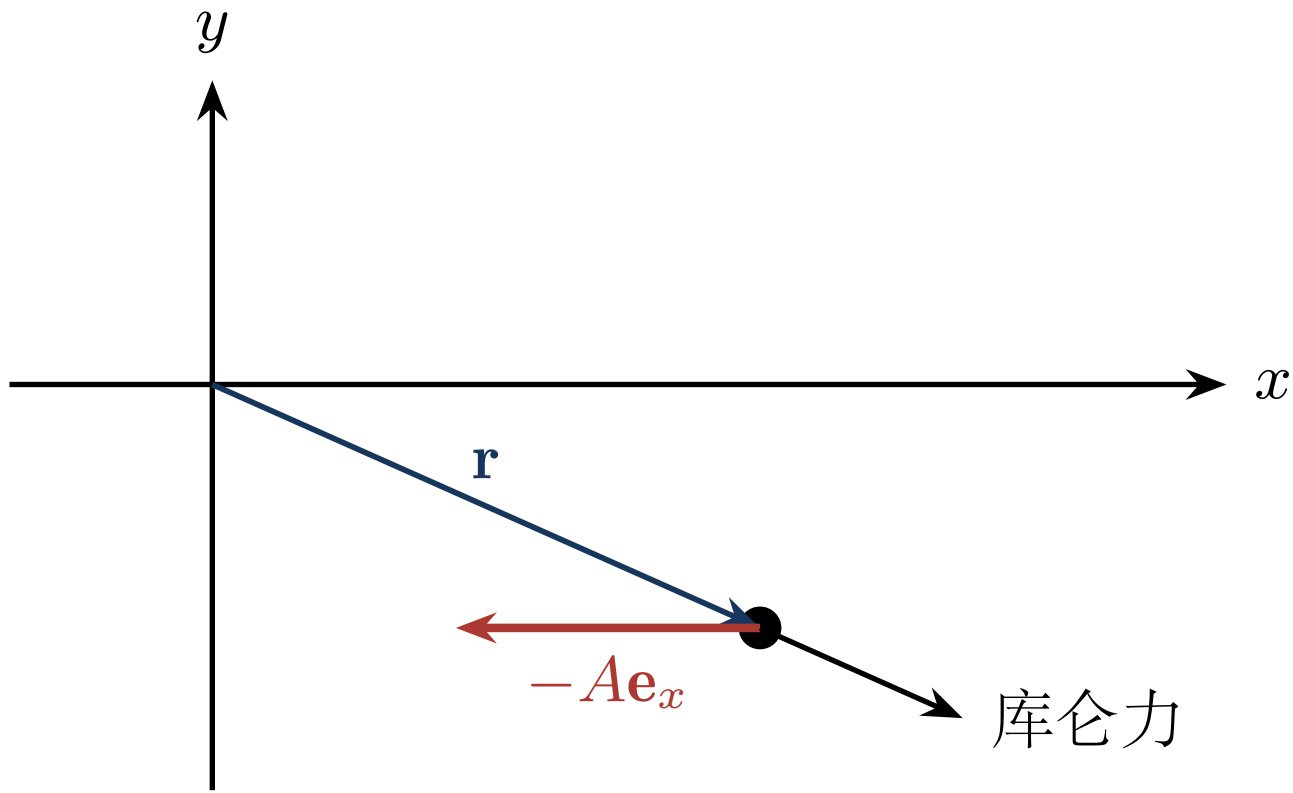
相对坐标 $\mathbf{r} = \mathbf{r}_M - \mathbf{r}_m$

初相对速度为 $(0, -v_0)$



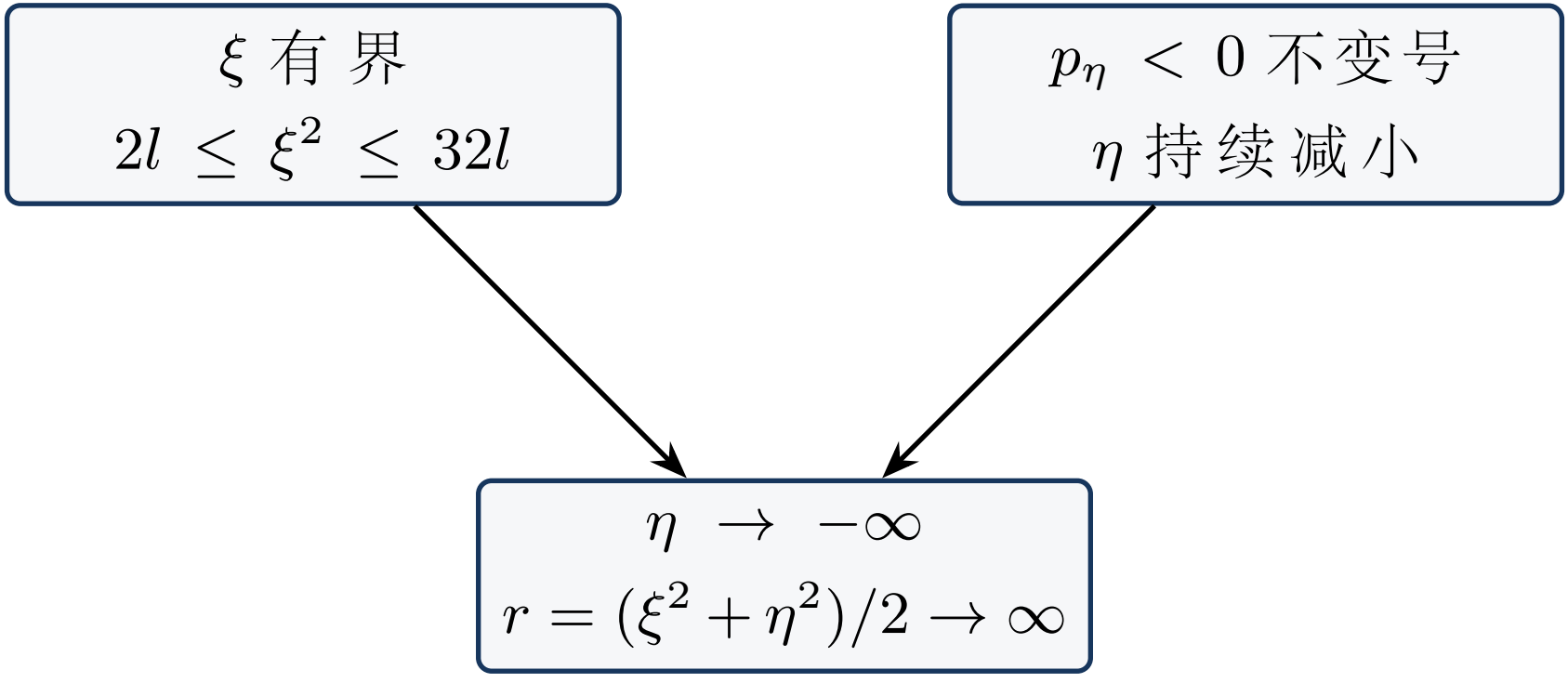
一维：$U = \kappa/r + Ar$

二维：$U = \kappa/r + Ax$，沿负 x 有逃逸通道



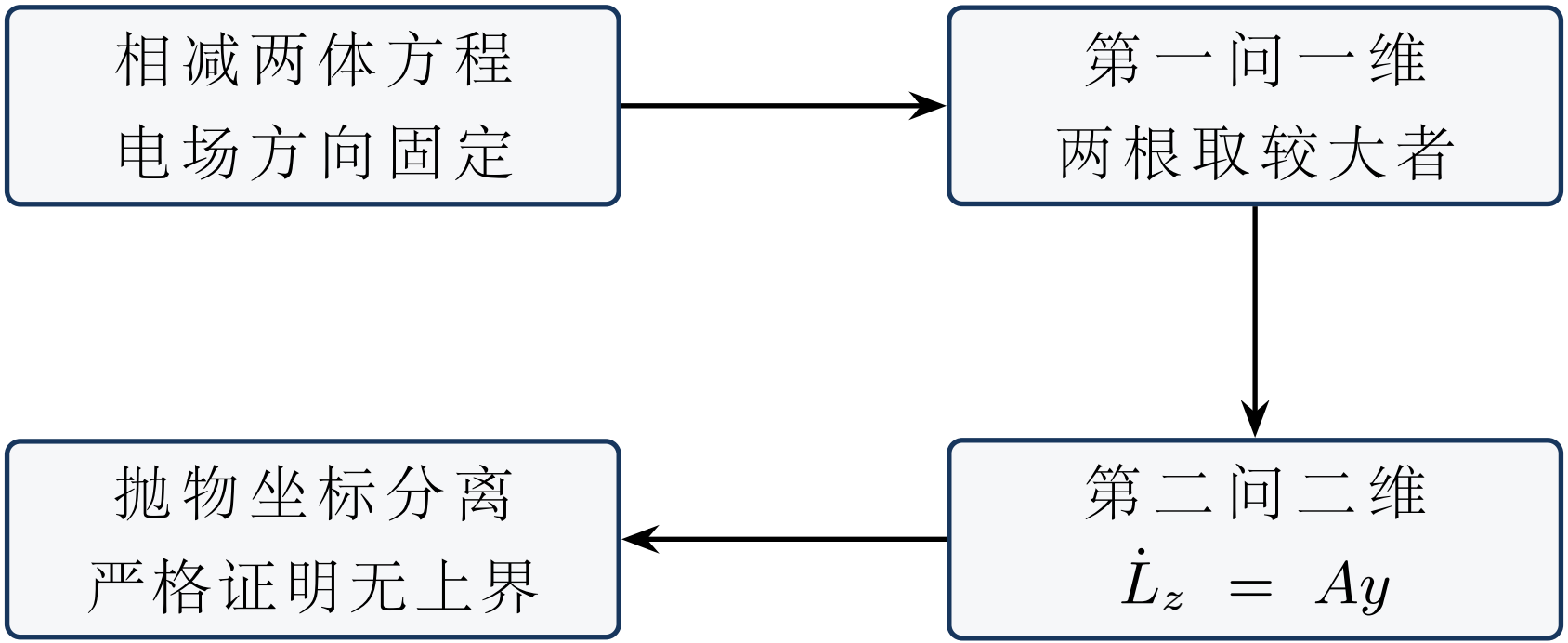
$\dot{L}_z = Ay$：固定方向外力有力矩

相对角动量不能当作常量



$$p_\eta^2/(2\mu) = A\eta^4/2 + H\eta^2 + 2\kappa/3 > 0$$

有限 η 区间内的 $|\dot{\eta}|$ 有正下界



$2l(4+\sqrt{17})$ 属于另加径向力的模型，不适用于原题